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LIQUID FILLING APPARATUS

Abstract

A liquid filling apparatus that fills a filled object with a liquid, the liquid filling apparatus includes: a liquid supply portion; a nozzle that includes a supply port and a sealing surface; a driving rod arranged inside a liquid supply channel so as to be freely movable in a vertical direction; and a sealing member fixed to the driving rod, wherein a gap is provided between a support surface of the driving rod and a facing surface of the sealing member.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a liquid filling apparatus that fills a filled object with a liquid.

BACKGROUND

[0002] In liquid blow molding apparatuses that manufacture synthetic resin containers by liquid blow molding, in which liquids are used as pressurizing media for blow molding, liquid filling apparatuses are used to fill (supply) pressurized liquids into preforms that are objects to be filled.

[0003] For example, Patent Literature (PTL) 1 describes a liquid filling apparatus including: a liquid supply portion configured to supply a liquid into a liquid supply channel; a nozzle that includes a supply port connected to the liquid supply channel and a sealing surface provided on an upstream side of the supply port; a driving rod arranged inside the liquid supply channel so as to be freely movable in a vertical direction; and a sealing member that has an abutting surface facing the sealing surface and that is fixed to the driving rod, wherein the abutting surface of the sealing member abuts the sealing surface to thereby close the supply port, and during liquid blow molding, the driving rod moves upward, and the abutting surface of the sealing member moves away from the sealing surface to thereby open the supply port, so that a pressurized liquid is supplied into a preform.

CITATION LIST

Patent Literature

[0004] PTL 1: JP 2019-130799 A

SUMMARY

Technical Problem

[0005] In the aforementioned conventional liquid filling apparatus, the inside of the liquid supply channel is always filled with the liquid in order to perform filling while pressurizing the liquid, so it is necessary to seal the supply channel with the sealing member. It is therefore preferable to use a synthetic resin material as the sealing member in order to improve sealing performance of the supply port by the sealing member.

[0006] With a configuration in which a synthetic resin material is used as the sealing member, however, in a case in which hot water is supplied into the liquid supply channel for cleaning or the like, or in a case in which a high-temperature liquid is filled into the filled object, for example, when such a high-temperature liquid is supplied into the liquid supply channel, the abutting surface of the sealing member is affected by heat and the sealing performance deteriorates, possibly resulting in leakage of the liquid from the supply port. When the liquid leaks from the supply port, the mold and the container as a product are spoiled by the leaked liquid.

[0007] To address this, it is possible to make the sealing member from metal, instead of synthetic resin. When the sealing member is made of metal, however, it is difficult to ensure sealing performance, and there is a risk of metal powder being mixed in with the liquid.

[0008] It would be helpful to provide a liquid filling apparatus capable of ensuring sealing performance with a synthetic resin sealing member, even when a high-temperature liquid is supplied into the liquid supply channel.

Solution to Problem

[0009] A liquid filling apparatus according to the present disclosure is a liquid filling apparatus that fills a filled object with a liquid, the liquid filling apparatus including: [0010] a liquid supply portion configured to supply the liquid into a liquid supply channel; [0011] a nozzle that includes a supply port connected to the liquid supply channel and a sealing surface provided on an upstream side of the supply port; [0012] a driving rod that includes a support surface facing the nozzle and an

engaged portion provided on the support surface and that is arranged inside the liquid supply channel so as to be freely movable in a vertical direction; and [0013] a synthetic resin sealing member, made of synthetic resin, that includes an abutting surface facing the sealing surface, a facing surface facing the support surface and an engaging portion provided on the facing surface, and that is configured to move along with the driving rod, by fixed to the driving rod by the engaging portion engages with the engaged portion, between a closed position, in which the abutting surface abuts the sealing surface to thereby close the supply port, and an open position, in which the abutting surface moves away from the sealing surface to thereby open the supply port, wherein [0014] a gap is provided between the support surface and the facing surface.
[0015] In a preferred embodiment of the liquid filling apparatus configured as above, [0016] the engaging portion includes a male thread portion protruding from the facing surface, [0017] the engaged portion includes a threaded hole, and [0018] a dimension in an axial direction of the engaging portion is set to be larger than a dimension in the axial dimension of the engaged portion.
[0019] In another preferred embodiment of the liquid filling apparatus configured as above, [0020] the engaging portion includes a male thread portion protruding from the facing surface, [0021] the engaged portion includes a threaded hole, and [0022] a dimension in an axial direction of the engaging portion is set to be equal to a dimension in the axial dimension of the engaged portion, and a spacer member is arranged between an end surface of the engaging portion and a bottom surface of the engaged portion.
[0023] In another preferred embodiment of the liquid filling apparatus configured as above, the supply port is configured as a cylindrical shape extending downward from an inner circumferential edge of the sealing surface.
[0024] In another preferred embodiment of the liquid filling apparatus configured as above, the liquid filling apparatus further includes: [0025] an extended portion that is integrally provided at a lower end of the sealing member and that is arranged inside the supply port when the sealing member is in the closed position; [0026] a draw-in hole that is provided in the extended portion and that includes one end opening at a lower end of the extended portion and another end opening at an outer circumferential surface of the extended portion; [0027] a draw-in channel that is provided in the nozzle and that communicates with the another end opening of the draw-in hole when the sealing member is in the closed position; and [0028] a negative pressure supply source connected to the draw-in channel.
Advantageous Effect
[0029] According to the present disclosure, a liquid filling apparatus capable of ensuring sealing performance with a synthetic resin sealing member, even when a high-temperature liquid is supplied into the liquid supply channel is provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0030] In the accompanying drawings:
[0031] FIG. **1** is an enlarged sectional view illustrating part of a liquid filling apparatus according to an embodiment of the present disclosure;
[0032] FIG. **2** is an enlarged sectional view illustrating part of a sealing member and a driving rod of FIG. **1**;
[0033] FIG. **3** is a bottom view illustrating the sealing member of FIG. **2**;
[0034] FIG. **4** is a sectional view illustrating the liquid filling apparatus of FIG. **1** in a state in which the sealing member is in an open position and a filled object is being filled with a liquid;
[0035] FIG. **5** is a sectional view illustrating the liquid filling apparatus of FIG. **1** in a state in which the sealing member is in a closed position after the filled object is filled with the liquid;

[0036] FIG. **6** is an enlarged sectional view illustrating part of a sealing member and a driving rod according to a modification; and
[0037] FIG. **7** is an enlarged sectional view illustrating part of a sealing member and a driving rod according to another modification.
DETAILED DESCRIPTION
[0038] In the following, a liquid filling apparatus according to an embodiment of the present disclosure will be more concretely described as illustration with reference to the drawings.
[0039] The liquid filling apparatus **1** of FIG. **1** according to the embodiment of the present disclosure constitutes a liquid blow molding apparatus that blow molds a filled object **2** into a container of a predetermined shape, by filling (supplying) a pressurized liquid **3** into the filled object **2**, which is a synthetic resin preform.
[0040] In the present embodiment, the filled object **2** is a preform that has been formed into a substantially test tube shape with a mouth **2***a* and a bottomed cylindrical body **2***b* using a synthetic resin material, such a polypropylene (PP) or polyethylene terephthalate (PET). The filled object **2** is heated to a predetermined temperature and placed in a blow molding mold **4** arranged below the liquid filling apparatus **1**, with the mouth **2***a* opening upward.
[0041] Additionally, the filled object **2** may be a preform of other materials and shapes.
[0042] The liquid filling apparatus **1** includes a main body block **10**. In the present embodiment, the main body block **10** includes a liquid supply channel **11** extending in a vertical direction inside the main body block **10**. A liquid supply portion **13** is connected to the liquid supply channel **11** via a pipe **12**. The liquid supply portion **13** is configured by, for example, a plunger pump, and it can supply the liquid **3** pressurized to a predetermined pressure into the liquid supply channel **11** through the pipe **12**. The liquid supply portion **13** can also supply a liquid (such as hot water) heated to a predetermined temperature into the liquid supply channel **11** through the pipe **12** when the liquid supply channel **11** is cleaned.
[0043] A nozzle **20** is attached to a lower end of the main body block **10**. The nozzle **20** includes a supply port **21**, which connects to the liquid supply channel **11** and which opens downward, and a sealing surface **22**, which is provided on an upstream side (upper side in FIG. **1**) of the supply port **21**.
[0044] In the present embodiment, the supply port **21** has a cylindrical shape extending downward from an inner circumferential edge of the sealing surface **22**, so as to be inserted inside the mouth **2***a* of the filled object **2**.
[0045] The supply port **21** is integrally provided, on the upper side thereof, with a holding portion **23** having a larger diameter than the supply port **21**. The nozzle **20** is fixed to the main body block **10**, by the holding portion **23** being held between a stepped surface **14** provided on an inner circumferential surface of the main body block **10** and a support block **15** fixed to a lower end of the main body block **10**.
[0046] The sealing surface **22** is provided as an upper surface of the holding portion **23**, and it is a tapered surface that gradually decreases in diameter from upward to downward. Additionally, the sealing surface **22** is not limited to a tapered surface.
[0047] The liquid filling apparatus **1** includes a sealing body **30** that opens and closes the supply port **21** of the nozzle **20**. The sealing body **30** includes a driving rod **31** and a sealing member **32**.
[0048] The driving rod **31** is arranged inside the liquid supply channel **11**, and it extends in the vertical direction along an axis O of the liquid supply channel **11**. The driving rod **31** is driven by a drive source that is not illustrated and is freely movable in the vertical direction so as to move closer to and away from the nozzle **20**. The driving rod **31** is integrally provided, at a lower end portion thereof, with a connecting portion **31***a* having a larger diameter than the upper portion. As illustrated in FIG. **1** and FIG. **2**, a lower surface of the connecting portion **31***a* that faces the nozzle **20** is a flat support surface **31***b* that is perpendicular to the axis O. The support surface **31***b* is provided with a threaded hole **31***c* as an engaged portion that is recessed at a predetermined depth

coaxially with the axis O.

[0049] The sealing member **32** is made of synthetic resin, and it includes an abutting surface **32***a* facing the sealing surface **22**, a facing surface **32***b* facing the support surface **31***b* of the driving rod **31**, and a male thread portion **32***c* as an engaging portion protruding from the facing surface **32***b* at a predetermined height coaxially with the axis O. In the present embodiment, the abutting surface **32***a* is provided as a lower surface of the sealing member **32** and is a tapered surface with a shape corresponding to the sealing surface **22**.

[0050] The sealing member **32** is fixed to the connecting portion **31***a* of the driving rod **31**, by the male thread portion **32***c* being screw-connected to the threaded hole **31***c* provided in the connecting portion **31***a* of the driving rod **31**. This makes the sealing member **32** freely movable along with the driving rod **31** between a closed position (position illustrated in FIG. **1**), in which the abutting surface **32***a* abuts the sealing surface **22** to thereby close the supply port **21**, and an open position (position illustrated in FIG. **6**), in which the abutting surface **32***a* moves away from the sealing surface **22** to thereby open the supply port **21**.

[0051] The sealing member **32** can also be configured to be integrally provided, at a lower end thereof, with an extended portion **33**.

[0052] As illustrated in FIG. **2**, the extended portion **33** has a cylindrical shape with substantially the same diameter as and coaxial with an inner circumferential surface of the supply port **21**, and it extends downward from the inner circumferential edge of the abutting surface **32***a*. The extended portion **33** is arranged inside the supply port **21** when the sealing member **32** is in the closed position. At this time, an outer circumferential surface of the extended portion **33** is in sliding contact or adjacent to the inner circumferential surface of the supply port **21**, and an end surface on the downstream side (lower side in FIG. **1**) of the extended portion **33** is arranged flush with an end surface on the downstream side of the supply port **21**.

[0053] The extended portion **33** includes draw-in holes **33***a*. Each draw-in hole **33***a* is an L-shaped flow channel with one end opening at a lower end of the extended portion **33** and another end opening at the outer circumferential surface of the extended portion **33**. In the present embodiment, the extended portion **33** is provided with a plurality of draw-in holes **33***a* arranged at regular intervals in the circumferential direction around the axis O. Accordingly, as illustrated in FIG. **3**, the respective one ends of the plurality of draw-in holes **33***a* are open on a lower end surface of the extended portion **33**, while being arranged at regular intervals in the circumferential direction. For convenience, only one draw-in hole **33***a* is denoted with a reference numeral in FIG. **3**.

[0054] As illustrated in FIG. **1**, in a case in which draw-in holes **33***a* are provided in the extended portion **33**, a draw-in channel **24** is provided in the nozzle **20**. The draw-in channel **24** is provided as an annular groove centered about the axis O on an inner circumferential surface of the holding portion **23** of the nozzle **20**, and it is configured to communicate with the respective other ends of the draw-in holes **33***a* when the sealing member **32** is in the closed position.

[0055] A negative pressure supply source **41** is connected to the draw-in channel **24** via a pipe **40**. The negative pressure supply source **41** is, for example, a vacuum pump. The pipe **40** is also provided with an opening and closing valve **42**. The opening and closing valve **42** is configured, for example, by a solenoid valve, and it is controlled to open and close by a control means that is not shown. When the opening and closing valve **42** is opened, the negative pressure supply source **41** communicates with the respective draw-in holes **33***a* via the pipe **40** and the draw-in channel **24**, so that negative pressure is supplied to the draw-in holes **33***a*.

[0056] The liquid filling apparatus **1** can also be configured to include a stretching rod **50**. In this case, the stretching rod **50** is arranged at an axis center of the driving rod **31** and the sealing member **32**, that is, the sealing body **30**, so as to be freely advance and retreat with respect to the sealing member **32**. During liquid blow molding when the pressurized liquid **3** is supplied into the filled object **2**, the stretching rod **50** is moved so that it protrudes downward from the sealing member **32**, and the filled object **2** is stretched longitudinally by the stretching rod **50**, and thus, the filled object **2** can be biaxially stretch blow molded.

[0057] Next, procedure for performing liquid blow molding by filling the filled object **2** with the pressurized liquid **3** using the liquid filling apparatus **1** will be described.

[0058] As illustrated in FIG. **1**, in a state in which the sealing member **32** is in the closed position, the supply port **21** is closed, and the opening and closing valve **42** is closed, the filled object **2**, which is an object to be filled with the liquid **3**, is placed in the mold **4** so that the mouth **2***a* is coaxial with the supply port **21**.

[0059] Subsequently, as illustrated in FIG. **4**, the main body block **10** is moved downward so that the supply port **21** of the nozzle **20** is fitted into the mouth **2***a* of the filled object **2**, and then the sealing body **30** is moved upward by operating the driving rod **31** with the opening and closing valve **42** being closed, to thereby move the sealing member **32** from the closed position to the open position. As a result, the abutting surface **32***a* moves away from the sealing surface **22**, the supply port **21** is opened, and the pressurized liquid **3** supplied from the liquid supply portion **13** to the nozzle **20** via the pipe **12** and the liquid supply channel **11** is filled from the supply port **21** into the filled object **2** through the mouth **2***a*, and thus, the body **2***b* of the filled object **2** is stretched. In the configuration in which the stretching rod **50** is provided, the body **2***b* of the filled object **2** is stretched longitudinally by the stretching rod **50** as the stretching rod **50** moves downward. When the filled object **2** is filled with a predetermined amount of the pressurized liquid **3**, the filled object **2** is formed into a container of a predetermined shape, and the filling of the liquid **3** into the filled object **2** is completed.

[0060] When the liquid **3** has been filled into the filled object **2**, as illustrated in FIG. **5**, the opening and closing valve **42** is opened with the negative pressure supply source **41** being operated, and the driving rod **31** is operated to move the sealing body **30** downward, to thereby move the sealing member **32** from the open position to the closed position. When the sealing member **32** is in the closed position, the abutting surface **32***a* abuts the sealing surface **22** over the entire circumference, so that the supply port **21** is closed, and the filling of the pressurized liquid **3** into the filled object **2** is stopped. The main body block **10** is then moved upward so as to separate the supply port **21** of the nozzle **20** from the mouth **2***a* of the filled object **2**. Additionally, the negative pressure supply source **41** may be left in a state of constant operation if it is being operated when the opening and closing valve **42** is opened, or it may start to operate when the opening and closing valve **42** is opened.

[0061] After the sealing member **32** is brought into the closed position and the supply port **21** is closed, the liquid **3** adhering to the end surface of the supply port **21** or the end surface of the extended portion **33** may drip, but because the opening and closing valve **42** is opened and negative pressure is supplied from the negative pressure supply source **41** to the plurality of draw-in holes **33***a* that open on the end surface of the extended portion **33**, the liquid **3** adhering to the end surface of the supply port **21** or the end surface of the extended portion **33** is drawn in through the plurality of draw-in holes **33***a*. This prevents liquid dripping from the end surface of the supply port **21** or the end surface of the extended portion **33**.

[0062] In the liquid filling apparatus **1** according to the present embodiment, the sealing member **32** is made of synthetic resin. For this reason, in a case in which hot water is supplied into the liquid supply channel **11** for cleaning or the like, or in a case in which a high-temperature liquid is filled into the filled object **2**, for example, when such a high-temperature liquid **3** is supplied into the liquid supply channel **11**, the abutting surface **32***a* of the sealing member **32** may be affected by heat, and the sealing performance may deteriorate.

[0063] To address this, in the liquid filling apparatus **1** according to the present embodiment, as illustrated in FIG. **2**, a gap **60** is provided between the support surface **31***b* of the driving rod **31** and the facing surface **32***b* of the sealing member **32**.

[0064] The gap **60** is provided outside the male thread portion **32***c* in an annular shape centered about the axis O and of a constant thickness. The gap **60** is set to be larger than or equal to the amount of elastic displacement in the vertical direction that occurs in the sealing member **32** when the sealing member **32** is held in the closed position under a predetermined load. The gap **60** provided between the support surface **31***b* and the facing surface **32***b* allows the sealing member **32** or the abutting surface **32***a* to be elastically deformed in a direction that narrows the gap **60** by the amount of the gap **60** when being held in the closed position with the predetermined load being applied from the driving rod **31**.

[0065] Thus, by providing the gap **60** between the support surface **31***b* of the driving rod **31** and the facing surface **32***b* of the sealing member **32**, even when the abutting surface **32***a* of the sealing member **32** is affected by heat, the sealing member **32** being held in the closed position by the predetermined load applied from the driving rod **31** can undergo further elastic deformation in the direction that narrows the gap **60**, thus providing adhesion of the abutting surface **32***a* to the sealing surface **22** and preventing liquid leakage. For example, even when deformation or distortion occurs in the abutting surface **32***a* due to heat, the sealing member **32** can undergo further elastic deformation in the direction that narrows the gap **60**, while being held in the closed position by the predetermined load applied from the driving rod **31**, thereby absorbing the deformation or the distortion, and thus, the adhesion to the sealing surface **22** can be provided.

[0066] In the present embodiment, a dimension in an axial direction of the male thread portion **32***c* is set to be larger than a dimension in the axial direction of the threaded hole **31***c*, and a gap **60** with a thickness corresponding to the difference in the dimensions is provided between the support surface **31***b* and the facing surface **32***b*.

[0067] According to this configuration, the gap **60** with a predetermined thickness can be easily and securely provided between the support surface **31***b* and the facing surface **32***b*, by screwing the male thread portion **32***c* into the threaded hole **31***c* with a predetermined torque until an end surface (upper end surface) of the male thread portion **32***c* abuts a bottom surface of the threaded hole **31***c*.

[0068] As illustrated in FIG. **6**, the dimension in the axial direction of the male thread portion **32***c* may be set to be equal to the dimension in the axial direction of the threaded hole **31***c*, and a spacer **61** with a predetermined thickness that is formed for example of stainless steel may be arranged between the end surface (upper end surface) of the male thread portion **32***c* and the bottom surface of the threaded hole **31***c*. By doing so, a gap **60** with a thickness corresponding to the thickness of the spacer **61** may also be configured to be provided between the support surface **31***b* and the facing surface **32***b*.

[0069] According to this configuration, in contrast to the sealing member **32** and the driving rod **31** of a conventional liquid filling apparatus in which a gap **60** is not provided between the support surface **31***b* and the facing surface **32***b*, the gap **60** can be easily and inexpensively provided between the support surface **31***b* and the facing surface **32***b*, by arranging the spacer **61** between the end surface of the male thread portion **32***c* and the bottom surface of the threaded hole **31***c*, while the conventional sealing member **32** and driving rod **31** are used.

[0070] As illustrated in FIG. **7**, the support surface **31***b* can be configured to be integrally provided, at an outer circumferential edge thereof, with a stopper portion **31***d*. The stopper portion **31***d* has an annular shape surrounding an outer circumference of the sealing member **32**, and its lower surface is a tapered surface that is offset upward in a stepped manner with respect to the abutting surface **32***a*. Additionally, there may be a radial gap between an inner circumferential surface of the stopper portion **31***d* and an outer circumferential surface of the sealing member **32**, or the inner circumferential surface of the stopper portion **31***d* and the outer circumferential surface of the sealing member **32** may abut each other.

[0071] According to this configuration, even when an excessive load is unexpectedly applied to the sealing member **32** from the driving rod **31**, the stopper portion **31***d* can abut the sealing surface **22**, thereby preventing the sealing member **32** from being damaged due to excessive elastic deformation.

[0072] Needless to say, the present disclosure is not limited to the above embodiment, and various

modifications can be made without departing from the gist of the present disclosure.
[0073] For example, although the above embodiment illustrates a case in which the liquid filling apparatus **1** constitutes a liquid blow molding apparatus used for liquid blow molding, the liquid filling apparatus **1** can also be used for filling a liquid as a content liquid into a container that has been formed in a predetermined shape in advance by blow molding or injection molding, without being limited to the case in which blow molding and filling are performed simultaneously.
[0074] Furthermore, although in the above embodiment the gap **60** has a constant thickness, the present disclosure is not limited to this. The thickness of the sealing member **32**, which can be elastically deformed by a desired amount of displacement, may not be constant.
[0075] Moreover, although in the above embodiment the supply port **21** is configured in a cylindrical shape extending downward from the inner circumferential edge of the sealing surface **22**, the present disclosure is not limited to this. The inner circumferential edge of the sealing surface **22** may be used as the supply port **21**, without providing the cylindrical portion.
[0076] Moreover, although in the above embodiment the threaded hole **31***c* is provided on the support surface **31***b* of the driving rod **31** as an engaged portion, the male thread portion **32***c* is provided on the facing surface **32***b* of the sealing member **32** as an engaging portion, and the sealing member **32** is fixed to the driving rod **31** by screw-connecting the male thread portion **32***c* to the threaded hole **31***c*, the present disclosure is not limited to this. A threaded hole as an engaging portion may be provided on the facing surface **32***b* of the sealing member **32**, and a male thread portion as an engaged portion that engages with the threaded hole may be provided on the support surface **31***b* of the driving rod **31**. The sealing member **32** may also be configured to be fixed to the driving rod **31** by means other than screw connection, by using configurations other than the male thread portion and the threaded hole as an engaging portion and an engaged portion.
REFERENCE SIGNS LIST
[0077] **1** Liquid filling apparatus [0078] **2** Filled object [0079] **2***a* Mouth [0080] **2***b* Body [0081] **3** Liquid [0082] **4** Mold [0083] **10** Main body block [0084] **11** Fluid supply channel [0085] **12** Pipe [0086] **13** Liquid supply channel [0087] **14** Stepped surface [0088] **15** Support block [0089] **20** Nozzle [0090] **21** Supply port [0091] **22** Sealing surface [0092] **23** Holding portion [0093] **24** Draw-in channel [0094] **30** Sealing body [0095] **31** Driving rod [0096] **31***a* Connecting portion [0097] **31***b* Support surface [0098] **31***c* Threaded hole (engaged portion) [0099] **31***d* Stopper portion [0100] **32** Sealing member [0101] **32***a* Abutting surface [0102] **32***b* Facing surface [0103] **32***c* Male thread portion (engaging portion) [0104] **33** Extended portion [0105] **33***a* Draw-in hole [0106] **40** Pipe [0107] **41** Negative pressure supply source [0108] **42** Opening and closing valve [0109] **50** Stretching rod [0110] **60** Gap [0111] **61** Spacer [0112] O Axis

Claims

1. A liquid filling apparatus that fills a filled object with a liquid, the liquid filling apparatus comprising: a liquid supply portion configured to supply the liquid into a liquid supply channel; a nozzle that includes a supply port connected to the liquid supply channel and a sealing surface provided on an upstream side of the supply port; a driving rod that includes a support surface facing the nozzle and an engaged portion provided on the support surface and that is arranged inside the liquid supply channel so as to be freely movable in a vertical direction; and a synthetic resin sealing member that includes an abutting surface facing the sealing surface, a facing surface facing the support surface, and an engaging portion provided on the facing surface and that is configured to move along with the driving rod between a closed position, in which the engaging portion engages with the engaged portion and is fixed to the driving rod, and the abutting surface abuts the sealing surface to thereby close the supply port, and an open position, in which the abutting surface moves away from the sealing surface to thereby open the supply port, wherein a gap is provided between the support surface and the facing surface.

2. The liquid filling apparatus according to claim 1, wherein the engaging portion comprises a male thread portion protruding from the facing surface, the engaged portion comprises a threaded hole, and a dimension in an axial direction of the engaging portion is set to be larger than a dimension in the axial dimension of the engaged portion.
3. The liquid filling apparatus according to claim 1, wherein the engaging portion comprises a male thread portion protruding from the facing surface, the engaged portion comprises a threaded hole, and a dimension in an axial direction of the engaging portion is set to be equal to a dimension in the axial dimension of the engaged portion, and a spacer member is arranged between an end surface of the engaging portion and a bottom surface of the engaged portion.
4. The liquid filling apparatus according to claim 1, wherein the supply port is configured as a cylindrical shape extending downward from an inner circumferential edge of the sealing surface.
5. The liquid filling apparatus according to claim 4, further comprising: an extended portion that is integrally provided at a lower end of the sealing member and that is arranged inside the supply port when the sealing member is in the closed position; a draw-in hole that is provided in the extended portion and that includes one end opening at a lower end of the extended portion and another end opening at an outer circumferential surface of the extended portion; a draw-in channel that is provided in the nozzle and that communicates with the other end of the draw-in hole when the sealing member is in the closed position; and a negative pressure supply source connected to the draw-in channel.
6. The liquid filling apparatus according to claim 2, wherein the supply port is configured as a cylindrical shape extending downward from an inner circumferential edge of the sealing surface.
7. The liquid filling apparatus according to claim 3, wherein the supply port is configured as a cylindrical shape extending downward from an inner circumferential edge of the sealing surface.
8. The liquid filling apparatus according to claim 6, further comprising: an extended portion that is integrally provided at a lower end of the sealing member and that is arranged inside the supply port when the sealing member is in the closed position; a draw-in hole that is provided in the extended portion and that includes one end opening at a lower end of the extended portion and another end opening at an outer circumferential surface of the extended portion; a draw-in channel that is provided in the nozzle and that communicates with the other end of the draw-in hole when the sealing member is in the closed position; and a negative pressure supply source connected to the draw-in channel.
9. The liquid filling apparatus according to claim 7, further comprising: an extended portion that is integrally provided at a lower end of the sealing member and that is arranged inside the supply port when the sealing member is in the closed position; a draw-in hole that is provided in the extended portion and that includes one end opening at a lower end of the extended portion and another end opening at an outer circumferential surface of the extended portion; a draw-in channel that is provided in the nozzle and that communicates with the other end of the draw-in hole when the sealing member is in the closed position; and a negative pressure supply source connected to the draw-in channel.